
Auditing VLM Benchmarks from Existing Evaluation Runs

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Abstract

1 Every evaluation run of a vision-language model (VLM) already produces per-
2 question artifacts (predictions, judged correctness, metadata) that leaderboards
3 reduce to a single score and then discard. We reuse those artifacts to audit the
4 benchmark itself, and add an audit layer to VLMEvalKit: six detectors that run
5 as post-processing over saved outputs, plus one additional text-only rerun with
6 visual inputs removed, emitting a per-question report in a single command. The
7 layer reports per-question findings together with aggregate audit statistics; for the
8 headline quantities, it attaches permutation-null excess where applicable, interval
9 estimates for proportions, and sensitivity to the model pool. Auditing MMStar,
10 a benchmark curated specifically so that the image is indispensable, with three
11 models, we find zero aggregate accuracy gain from the image on 20.2% of its
12 questions. An item-matched permutation null accounts for 17.0%, leaving a
13 +3.2 pp excess (corrected permutation $p = 0.001$). This excess is concentrated in
14 the mathematics and science categories and is absent from coarse perception, a
15 localization consistent with knowledge substituting for visual evidence; confirming
16 that mechanism would require item-level review. A second detector flags 134
17 questions on which every model that answered in parseable form picked the same
18 non-gold option, and these candidates peak in the perception categories, whereas
19 the zero-gain excess is largest in mathematics and science & technology. The
20 contrasting category profiles show that the two detectors capture distinct signals.
21 Replacing the LLM judge with exact string matching changes no conclusion,
22 and every scoring discrepancy between the two traces to one model whose free-
23 form answers a parser cannot read. On a binary yes/no benchmark the detectors
24 measurably fall out of calibration, a scope boundary that we measure and report.
25 The audit layer, the prediction artifacts and the generated reports accompany this
26 submission.

27 1 Introduction

28 Multimodal benchmarks are the instruments through which progress in vision-language models is
29 measured, compared and rewarded. The scores they produce steer model development and public
30 leaderboards, yet the benchmarks themselves are seldom measured with comparable care. A multiple-
31 choice suite can contain questions answerable without looking at the image, gold labels that are
32 wrong, and option positions that reward a guessing strategy, and a model’s reported accuracy absorbs
33 all of these defects silently. Checking for them is usually treated as a separate research effort, either
34 by building a cleaner replacement benchmark or by running a dedicated flaw-mining pipeline, each
35 with its own annotation budget and inference bill, so in practice such checks happen rarely and cover
36 few benchmarks. Yet every evaluation run already produces the raw material for exactly this kind of
37 check: per-question predictions, judged correctness and metadata that leaderboards show as a single

38 score. This paper asks how far that discarded material can be pushed as an audit of the benchmark
39 itself.

40 Our case study is MMStar, a benchmark built by filtering candidate questions so that the image is
41 essential: items solvable by blind language models were removed by construction. We run three
42 VLMs on it twice, first as intended and then with every image removed from the prompt, and align
43 the two runs question by question. On 303 of 1,500 questions (20.2%) the pool is exactly as accurate
44 without the image as with it. Most of that share is plain arithmetic: even blind answers unrelated to
45 the items sometimes tie full-run accuracy, and an item-matched permutation null puts that baseline at
46 17.0%. The remaining +3.2 pp ($p = 0.001$) is a paired zero-gain signal, and it concentrates in the
47 two categories where non-visual knowledge is a plausible explanation. Finding this required no new
48 benchmark and no annotation campaign. The audit reuses the full-run artifacts and adds one text-only
49 inference pass per model; this pass avoids visual processing but is not cost-free.

50 We implement this idea as an audit layer inside a mainstream evaluation toolkit. Six detectors run
51 as post-processing over the per-question files the toolkit already writes; the only added inference
52 is the text-only rerun. The layer pairs each headline quantity with the statistics needed to read it
53 conservatively, and the case study is designed to probe the audit’s own blind spots as well as the
54 benchmark’s.

55 1.1 Contributions

- 56 1. **The first benchmark-audit layer inside a mainstream VLM evaluation toolkit.** We
57 extend VLMEvalKit with six benchmark-pathology detectors and a toolkit-wide text-only
58 blind-run mode; a single command turns the saved artifacts of any completed evaluation into
59 a per-question audit report with uncertainty statistics and a reproducibility manifest. To our
60 knowledge, no mainstream VLM evaluation toolkit previously included benchmark-quality
61 checks (Section 3).
- 62 2. **A statistical protocol that separates audit signal from accuracy arithmetic.** We report
63 every headline quantity as an excess over an item-matched permutation null, with interval
64 estimates, leave-one-model-out ranges and category strata. On MMStar this protocol turns a
65 raw 20.2% zero-gain share into a defensible +3.2 pp excess ($p = 0.001$) and exposes how
66 unstable the raw share alone is: it varies by 14.2 pp across model pools while the excess
67 varies by 4.6 pp (Section 3.3).
- 68 3. **The first published item-level audit of MMStar, with quantified validity limits.** We
69 localize two distinct signals inside a benchmark curated for vision-indispensability: the zero-
70 gain excess concentrates in mathematics and science & technology, while 134 unanimous
71 label-conflict candidates concentrate in the perception categories, at a rate consistent with an
72 independent re-annotation study. The audit also measures its own limits: every conclusion
73 survives removal of the LLM judge, and binary yes/no formats fall measurably outside
74 calibration (Sections 4.1–4.5).

75 The pathologies these detectors target are documented at scale by prior and concurrent work (Sec-
76 tion 2); our claims concern the infrastructure, the statistical protocol and the item-level findings. One
77 measurement appears to be new outright: Section 4.4 quantifies how the answer-extraction protocol
78 shifts audit conclusions themselves, where prior work measured its effect on model accuracy.

79 2 Related work

80 **Benchmark pathologies.** Language priors in visual question answering were established by Goyal
81 et al. [2017]; MMStar [Chen et al., 2024] and MMEvalPro [Huang et al., 2024] showed that blind
82 language models score far above chance on multimodal multiple-choice suites; concurrent work
83 probes the visual dependence of existing suites through blur ablations [Zhou et al., 2026] or phrasing-
84 controlled items [Singla et al., 2026]; HallusionBench [Guan et al., 2024] builds yes/no items around
85 visual illusions and language priors to elicit these failure modes deliberately. Label errors destabilize
86 rankings [Northcutt et al., 2021]; re-annotation efforts include MMLU-Redux [Gema et al., 2025]
87 and platinum benchmarks [Vendrow et al., 2025], and Li et al. [2025] report a fallibility analysis
88 covering MMStar. Position and selection bias in multiple choice is established for language models
89 [Zheng et al., 2024], handled in MMBench through CircularEval [Liu et al., 2024], and reused as a

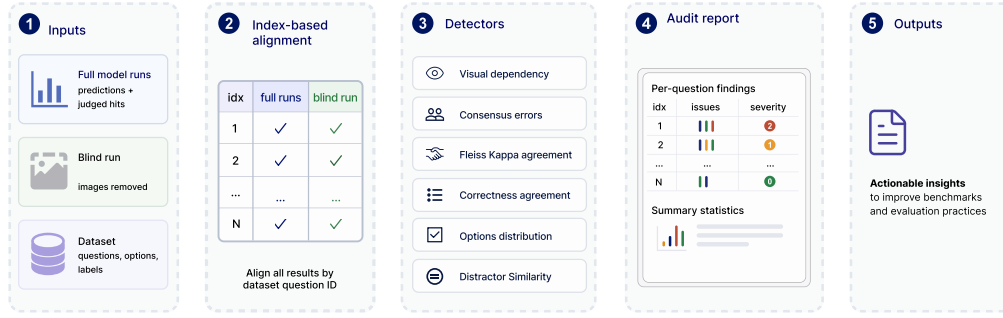


Figure 1: The audit pipeline in five stages. **(1)** Inputs: the full model runs with predictions and judged hits, one additional text-only blind run with images removed, and the dataset itself with questions, options and gold labels. **(2)** All frames are aligned by dataset question ID. **(3)** Six detectors process the aligned frame. **(4)** The audit report lists per-question findings with severity levels and summary statistics, including permutation-null excess values and a reproducibility manifest. **(5)** The report yields actionable insights for improving benchmarks and evaluation practice. The blind run is the only step that requires inference, and it runs without images.

contamination probe [Song et al., 2024]. Kim et al. [2025] show that model errors correlate across vendors and scales, which caps what any cross-model consensus can establish; we return to this in Sections 4.3 and 5.

Audit pipelines and toolkits. The standard responses to these pathologies treat auditing as a separate research project with its own inference budget, either by building cleaner replacement benchmarks [Chen et al., 2024, Li et al., 2024, Vendrow et al., 2025, Joshi et al., 2026] or by running standalone flaw-mining pipelines. Fantastic Bugs [Truong et al., 2025] mines response patterns to flag invalid questions in nine text benchmarks; BenchMarker [Balepur et al., 2026] applies an education-inspired rubric with LLM judges to text multiple-choice benchmarks; DatBench [Joshi et al., 2026] audits 33 VLM datasets and releases a cleaned suite. Our deliverable differs: audit machinery embedded in a mainstream toolkit, reusable on any benchmark-model combination from artifacts the toolkit already saves. To our knowledge no such layer exists for VLMEvalKit [Duan et al., 2024] or LMMs-Eval [Zhang et al., 2024]. Blind-solvability and label errors in VLM benchmarks are documented at scale by concurrent work [Joshi et al., 2026, Zhou et al., 2026]; the checks themselves are therefore established, and this paper moves them inside the evaluation infrastructure. Redundancy analyses over existing score matrices [Zhang et al., 2025] share the reuse-the-exhaust spirit for a different question. On the measurement side, Rosenthal et al. [2025] document protocol sensitivity of VQA accuracy; we extend that question from accuracy numbers to audit conclusions.

3 Methodology

3.1 Inputs and the blind-run protocol

Figure 1 gives an overview of the pipeline. The layer consumes the per-question result files of n models on one benchmark, each with a raw prediction, the gold answer, and optionally a judged correctness flag. A *blind run* repeats inference with visual content removed and textual content unchanged. The standard local and API wrapper paths share the same stripping function; blind artifacts get a `_blind` suffix and sit beside the full run. The toolkit audit pipeline aligns frames on the dataset-level question ID and restricts them to the shared ID set.

Two rules protect validity. First, model answers are never back-filled from gold labels, whatever the judge decided; an earlier version of our own pipeline made exactly this mistake, and repairing it multiplied the unanimous-conflict count by eleven (Section 4.6). Second, predictions from which no option letter can be parsed become an abstention sentinel that is excluded from consensus votes instead of being silently counted as wrong.

Table 1: Detector taxonomy. “Judge-dep.” marks detectors whose output depends on the answer-extraction protocol.

Detector	Signal	Pathology flagged	Judge-dep.
visual_dependency	full vs. blind accuracy	no-gain-from-image items	yes
consensus_error	majority vote vs. label	label-conflict candidates	no
fleiss_kappa_agreement	inter-model κ	unstable / ambiguous items	no
correctness_agreement	judged-correctness profile	difficulty / discrimination	yes
answer_options_distribution	label-position histogram	position bias	no
distractor_similarity	pairwise option similarity	near-duplicate options	no

3.2 Detectors

Table 1 summarizes the six detectors. Consensus, agreement, position-distribution and distractor detectors parse option letters from raw predictions and never consult a judge; visual dependency and correctness agreement consume judged labels and are re-checked under exact matching in Section 4.4.

For visual dependency, each question is categorized from model-averaged full accuracy a_q^F and blind accuracy a_q^B : *visual-dependent* ($a_q^F \geq 0.99$ and $a_q^B \leq 0.01$), *conflicting* ($a_q^B > a_q^F + \delta$, the image hurts), *visual supplement* ($a_q^F - a_q^B \geq \delta$), and *text-only* otherwise, with $\delta = 0.2$. With $n = 3$ models the accuracies live on the grid $\{0, 1/3, 2/3, 1\}$, so text-only reduces to exact equality $a_q^F = a_q^B$ and carries no tunable threshold. The visual-dependent share, in contrast, is threshold-fragile: it is 22.5% under the strict definition and 54.8% under a majority definition (Appendix B), which is why we never headline it. Consensus conflicts are tiered: *unanimous* (every parseable answer is the same non-gold option), *high* (all models disagree with the gold label), *medium* (a qualified majority disagrees).

3.3 Audit statistics based on the excess over a permutation null

A raw zero-gain share is easy to over-read. Even if blind answers were drawn independently of the items, some questions would tie full-run accuracy by chance, and that baseline moves with the pool’s marginal accuracy; across our two- and three-model pools the raw share varies by 14.2 pp. The layer therefore reports the share’s *excess over a permutation null*: blind score vectors are shuffled across questions, with the same permutation applied to every model, which preserves each model’s marginal blind accuracy and the inter-model correlation structure while destroying only the per-question full-to-blind pairing. We use 999 permutations; the one-sided p -value uses the standard $(g+1)/(N+1)$ correction, so the smallest reportable value is $p = 0.001$. The same module attaches Wilson 95% intervals to headline proportions (item-level uncertainty), leave-one-model-out pools (pool sensitivity), a split-half reproducibility check, and a stratification by official dataset categories (localization and internal replication). All of it is post-processing without any model use.

3.4 Evaluation protocol

Benchmarks. MMStar: 1,500 four-option multiple-choice items curated for vision-indispensability, with six official categories of 250 items each. HallusionBench: 951 yes/no items built around visual illusions and language priors, used here as a deliberate out-of-format probe of the detectors themselves.

Models. InternVL3-8B, Ristretto-3B and gpt-5-nano ($n = 3$): two open models and one API model. This case study is sized to validate the audit machinery; leave-one-model-out ranges are reported throughout so that pool sensitivity stays explicit (Table 2).

Scoring. The judged arm uses gpt-4o-mini as the answer-extraction judge, identical for full and blind runs; the exact arm parses option letters directly from predictions with no model in the loop. Consensus, κ , position and distractor detectors use the exact arm by construction. Both arms are reported wherever they can differ.

Reproducibility. Every run records its git commit, command-line arguments and generation parameters in a manifest; the audit is deterministic given the artifacts and a fixed permutation seed. The audit layer, the prediction artifacts for both benchmarks, and the generated reports accompany this

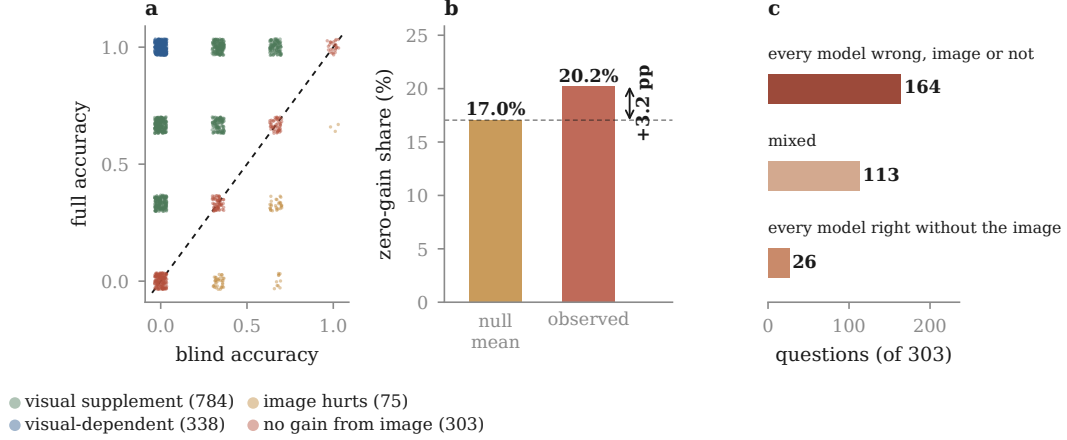


Figure 2: The zero-gain signal on MMStar. **(a)** Model-averaged full against blind accuracy per question (judged arm, jittered; three models give the 4×4 grid). Questions on the diagonal gain nothing from the image. **(b)** The observed 20.2% share against 999 item-shuffled permutations: the null centres at 17.0%, leaving a +3.2 pp excess that no permutation reaches ($p = 0.001$). **(c)** The no-gain set is heterogeneous: on 164 of 303 questions every model is wrong with and without the image, so the conservative reading is *zero aggregate accuracy gain from the image for this model pool*, and genuinely text-answerable items are only one possible component.

160 submission; Appendix A gives the exact commands that regenerate every number and figure in this
 161 paper.

162 4 Results

163 4.1 A curated benchmark still carries an excess zero-gain rate

164 Figure 2 and Table 2 give the headline result. The observed zero-gain share is 20.2% (Wilson 95% CI
 165 [18.2, 22.3]); the permutation null predicts 17.0%; the excess is +3.2 pp at $p = 0.001$, the floor of a
 166 999-permutation test. The excess survives every leave-one-model-out pool (+3.6 to +7.7 pp, each
 167 $p = 0.001$). Across 20 random split-half checks, the median absolute difference between the two
 168 half-sample excess estimates is 1.5 pp.

169 Two readings of this result must be kept apart (Figure 2c). Of the 303 zero-gain questions, 164 are
 170 items on which every model is wrong both with and without the image, which is evidence of shared
 171 failure. Only 26 are answered correctly by all three models blind. Our claim is therefore limited to
 172 equal model-averaged full and blind accuracy for this pool. The excess quantifies how often that
 173 equality occurs beyond marginal-rate arithmetic; it does not identify which underlying mechanism
 174 produced it. The modest +3.2 pp excess is also consistent with MMStar’s curation having removed
 175 most non-visual shortcuts.

176 4.2 The two audit signals have different category profiles

177 Stratifying by MMStar’s six official categories turns one number into a pattern (Figure 3). The
 178 zero-gain excess is largest in math (+6.6 pp) and science & technology (+7.5 pp), where non-
 179 visual knowledge is a plausible explanation. Coarse perception shows no positive excess (−1.7 pp).
 180 Unanimous label-conflict candidates distribute differently: they peak in the perception categories
 181 (11.6% coarse, 14.4% fine-grained) and are lowest in math and reasoning (4.4–6.8%), with science &
 182 technology intermediate (10.4%). These category profiles show that the two detectors capture distinct
 183 signals; neither profile by itself identifies the causal mechanism behind its signal.

184 **Blind-run abstentions.** The extraction discrepancy is distinct from refusal under the blind ab-
 185 lation. Full/blind unparseable rates are 0.0/20.5% for InternVL3-8B, 0.0/1.1% for Ristretto-3B,
 186 and 10.3/67.5% for gpt-5-nano. Consequently, zero-gain and visual-gain categories reflect both

Table 2: Zero-gain share against the permutation null on MMStar (999 permutations, one-sided, $(g+1)/(N+1)$ correction). The excess is markedly more pool-stable than the raw share: raw spread 14.2 pp, excess spread 4.6 pp.

Model pool	Observed (%)	Null mean (%)	Excess (pp)	p
All three models	20.2	17.0	+3.2	0.001
without InternVL3-8B	27.9	24.1	+3.7	0.001
without Ristretto-3B	25.5	21.9	+3.6	0.001
without gpt-5-nano	34.4	26.7	+7.7	0.001

187 benchmark properties and model-specific willingness to answer without an image. We retain all three
 188 models in the primary pool and expose this dependence through the leave-one-model-out analysis;
 189 results should not be extrapolated to pools with different refusal policies.

190 4.3 Label-conflict candidates

191 The consensus detector flags 23.9% of MMStar items at some tier; we focus on the unanimous
 192 tier. On 134 questions (8.9%, Wilson 95% CI [7.6, 10.5]) every parseable model answer is the
 193 same option and it is not the gold label; on 117 of those all three models answered in parseable
 194 form, and on the remaining 17, only two models produced parseable answers because abstentions
 195 are excluded from the vote. On the 1,346 items for which all three models produced parseable
 196 option answers, complete-case Fleiss’ κ is 0.544 (item-bootstrap 95% CI [0.517, 0.570]), indicating
 197 moderate agreement beyond that expected from the models’ marginal answer distributions. Because
 198 the excluded 154 items are non-random unparseable outputs from gpt-5-nano, this estimate describes
 199 the complete-case subset rather than all 1,500 MMStar items. Unanimous agreement therefore stands
 200 well apart from the pool’s baseline behavior. The rate is of the same order as the annotation-error
 201 rates reported for MMStar by the independent fallibility analysis discussed in Section 2, which did
 202 collect fresh human labels.

203 We deliberately call these *candidates*, and the reason is measurable in our own pool. Errors of
 204 different models are correlated (Section 2): when two of our models are both wrong, they pick the
 205 same wrong option in 64–74% of cases, against 26–27% when the item-to-item pairing is destroyed
 206 by the same shuffle that defines our null. Unanimity therefore serves as a prioritization signal: it
 207 concentrates human review, which remains the arbiter. The audit layer releases the flagged list with
 208 per-question model votes for exactly that purpose.

209 4.4 Removing the LLM judge changes almost nothing, for a reason

210 Audit claims should not silently inherit an LLM judge. Four of six detectors never use one (Table 1).
 211 For the two that do, we recomputed everything under exact letter matching on identical predictions
 212 (Figure 4a): category shares move by at most 2.1 pp (zero-gain 19.8% against 20.2%; visual-dependent
 213 20.5% against 22.5%; supplement 53.6% against 52.3%; conflicting 6.1% against 5.0%), and nothing
 214 in Sections 4.1–4.3 changes.

215 The instructive part is where the two arms disagree. 122 of 1,500 items (8.1%) change visual-
 216 dependency category between arms, and every one of them traces to gpt-5-nano: the judge and the
 217 letter parser score that model differently on 229 model-item pairs, and the two open models on none
 218 at all (Figure 4b). The mechanism is prosaic: gpt-5-nano answers in free-form prose on 10% of
 219 items, which a regular expression cannot read but a judge can. The judge is thus a robustness arm:
 220 for models with canonical outputs exact matching suffices, and for models without canonical outputs
 221 the audit report can say so per model instead of averaging the discrepancy away.

222 4.5 Where the detectors stop working: a binary-format probe

223 We ran the same audit on HallusionBench, whose 951 items are yes/no questions, to measure how the
 224 detectors behave outside their design format. With two answer options blind guessing sits at 50%,
 225 unanimity is easy to obtain by chance, and the multiple-choice consensus machinery does not apply at
 226 all: option letters do not parse and the exact arm reports abstention everywhere. The zero-gain excess
 227 is still large and significant (+5.4 pp over an 18.5% null, $p = 0.001$), but the decisive calibration test

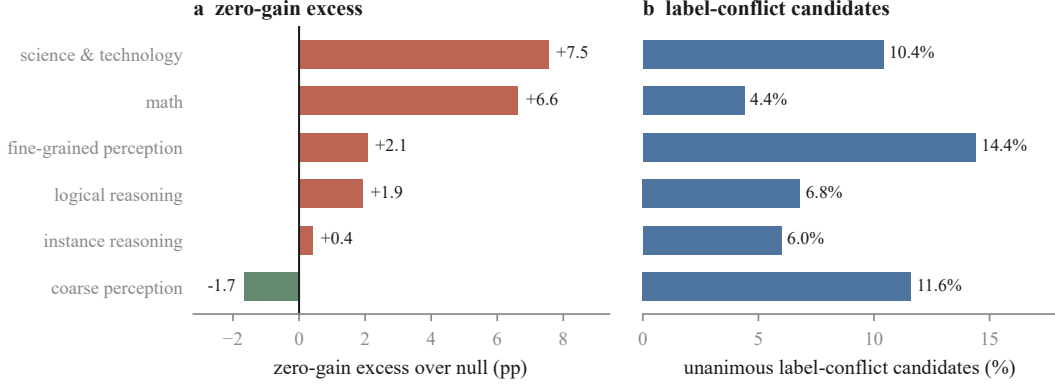


Figure 3: Per-category audit of MMStar (250 items per category), sorted by zero-gain excess. (a) The excess concentrates in mathematics and science & technology and is absent from coarse perception, a pattern consistent with greater scope for non-visual knowledge in the former categories. (b) Unanimous label-conflict candidates peak in the perception categories. The two detectors localize different candidate failure modes.

fails. Ranking items by their continuous visual gain recovers HallusionBench’s own visual-dependent against visual-supplement design labels at AUC 0.49, that is, at chance (Appendix C). One directional signal survives and matches the benchmark’s intent: the excess is approximately 2.2 times larger on design-VS items (+7.6 pp) than on design-VD items (+3.4 pp). We conclude that the current evidence supports use on the four-option MMStar setting evaluated here and places binary formats outside the demonstrated scope. Calibration across other k -way multiple-choice benchmarks remains to be measured.

4.6 Auditing the audit

Audit pipelines can manufacture their own findings, so we checked ours against the known failure modes. *No-op perturbation*: the blind run strips images centrally, blind transcripts contain model statements that no image is present, and blind accuracy falls towards the option prior in the perception categories. *Extraction artifacts*: the judged-against-exact arm (Section 4.4) measures precisely this and attributes all discrepancy to one model. *Unfaithful scoring*: the judged arm is VLMEvalKit’s native protocol, unmodified. *Broken pairing in uncertainty*: the permutation null is item-paired by construction, and only the full-to-blind linkage is destroyed. *One scalar, two readings*: every headline is disaggregated (category strata, per-pool tables, and the 164-of-303 composition in Figure 2c). One bug we did catch and fix is reported in Section 3.1: back-filling gold labels into model answers had suppressed unanimous conflicts from 134 to 12, and the layer now refuses to do it.

5 Limitations

Model pool. A pool of three models fixes the audit’s signals but leaves the exact category shares pool-dependent; we treat it as a working approximation whose sensitivity is measured rather than assumed. The direction and significance of the headline excess are unchanged in every leave-one-model-out pool (Table 2), category shares are published with their pool ranges, and each model’s blind-refusal profile is reported alongside (Section 4.2). A larger pool will tighten the per-item accuracy grid and is part of ongoing work. *Mechanism attribution*. The zero-gain excess is an aggregate, mechanism-agnostic quantity: it bounds the combined effect of language priors, option statistics and contamination, and the category strata localize it without singling out one cause. *Scope*. We audit one benchmark in depth under a four-option multiple-choice format. The binary-format boundary is measured in Section 4.5, and base rates are benchmark-specific by construction, so applying the audit to a new suite means rerunning it there.

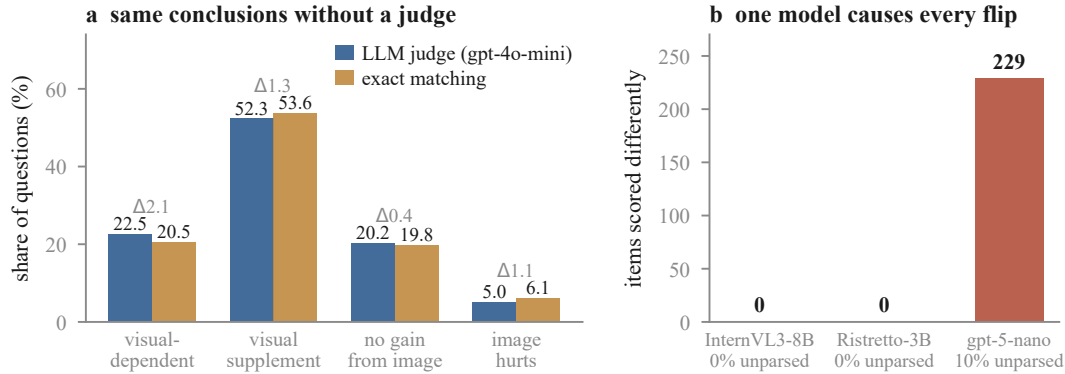


Figure 4: Judge dependence of the audit on MMStar. **(a)** Category shares under the gpt-4o-mini judge and under exact letter matching, on identical predictions: every share moves by at most 2.1 pp and no conclusion changes. **(b)** The disagreement has a single source: for the two open models the judge and the letter parser score every item identically, while gpt-5-nano, whose answers are unparseable on 10% of items, accounts for all of it.

6 Conclusion

An evaluation toolkit already stores what a benchmark audit needs; one text-only rerun and a statistics pass turn those artifacts into per-item findings with uncertainty attached. On MMStar the audit surfaces a +3.2 pp excess zero-gain rate concentrated in knowledge-heavy categories and 134 unanimous label-conflict candidates concentrated in perception categories, and its validity checks bound its own scope: the conclusions survive removal of the LLM judge, and binary yes/no formats fall outside the demonstrated calibration. The broader lesson is a reporting discipline: a headline claim about a benchmark’s defects should carry its permutation null, interval estimate, model-pool sensitivity and refusal profile, because a raw flag count invites over-reading. We are extending the pool to six models, running human verification of the flagged items, and sweeping the remaining benchmarks of the toolkit; the layer, artifacts and reports are released so that every evaluation run can double as a benchmark audit.

References

- Nishant Balepur, Bhavya Rajasekaran, Jane Oh, Michael Xie, Atrey Desai, Vipul Gupta, Steven James Moore, Eunsol Choi, Rachel Rudinger, and Jordan Lee Boyd-Graber. BenchMarker: An education-inspired toolkit for highlighting flaws in multiple-choice benchmarks. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026. URL <https://arxiv.org/abs/2602.06221>.
- Lin Chen, Jinsong Li, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Zehui Chen, Haodong Duan, Jiaqi Wang, Yu Qiao, Dahua Lin, and Feng Zhao. Are we on the right way for evaluating large vision-language models? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. URL <https://arxiv.org/abs/2403.20330>.
- Haodong Duan, Xinyu Fang, Junming Yang, Xiangyu Zhao, Zerun Ma, Yuxuan Qiao, Mo Li, Tianhao Liang, Lin Zhu, et al. VLMEvalKit: An open-source toolkit for evaluating large multi-modality models. In *Proceedings of the 32nd ACM International Conference on Multimedia (ACM MM)*, 2024. URL <https://arxiv.org/abs/2407.11691>.
- Aryo Pradipta Gema, Joshua Ong Jun Leang, Giwon Hong, Alessio Devoto, Alberto Carlo Maria Mancino, Rohit Saxena, Xuanli He, Yu Zhao, Xiaotang Du, Mohammad Reza Ghasemi Madani, Claire Barale, Robert McHardy, Joshua Harris, Jean Kaddour, Emile van Krieken, and Pasquale Minervini. Are we done with MMLU? In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL)*, 2025. URL <https://aclanthology.org/2025.naacl-long.262/>.
- Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. Making the V in VQA matter: Elevating the role of image understanding in visual question answering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017. URL <https://arxiv.org/abs/1612.00837>.

291 Tianrui Guan, Fuxiao Liu, Xiyang Wu, Ruiqi Xian, Zongxia Li, Xiaoyu Liu, Xijun Wang, Lichang Chen, Furong
292 Huang, Yaser Yacoob, Dinesh Manocha, and Tianyi Zhou. HallusionBench: An advanced diagnostic suite
293 for entangled language hallucination and visual illusion in large vision-language models. In *Proceedings*
294 *of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. URL <https://arxiv.org/abs/2310.14566>.
295

296 Jinsheng Huang, Liang Chen, Taian Guo, Fu Zeng, Yusheng Zhao, Bohan Wu, Ye Yuan, Haozhe Zhao, Zhihui
297 Guo, Yichi Zhang, Jingyang Yuan, Wei Ju, Luchen Liu, Tianyu Liu, Baobao Chang, and Ming Zhang.
298 MMEvalPro: Calibrating multimodal benchmarks towards trustworthy and efficient evaluation. *arXiv preprint*
299 *arXiv:2407.00468*, 2024. URL <https://arxiv.org/abs/2407.00468>.

300 Siddharth Joshi, Haoli Yin, Rishabh Adiga, et al. DatBench: Discriminative, faithful, and efficient VLM
301 evaluations. *arXiv preprint arXiv:2601.02316*, 2026. URL <https://arxiv.org/abs/2601.02316>.

302 Elliot Kim, Avi Garg, Kenny Peng, and Nikhil Garg. Correlated errors in large language models. In *International*
303 *Conference on Machine Learning (ICML)*, 2025. URL <https://arxiv.org/abs/2506.07962>.

304 Baiqi Li, Zhiqiu Lin, Wenxuan Peng, Jean de Dieu Nyandwi, Daniel Jiang, Zixian Ma, Simran Khanuja,
305 Ranjay Krishna, Graham Neubig, and Deva Ramanan. NaturalBench: Evaluating vision-language models on
306 natural adversarial samples. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. URL
307 <https://arxiv.org/abs/2410.14669>.

308 Chunyi Li, Xiaozhe Li, Zicheng Zhang, Yuan Tian, Ziheng Jia, Xiaohong Liu, Xionghuo Min, Jia Wang,
309 Haodong Duan, Kai Chen, and Guangtao Zhai. Information density principle for MLLM benchmarks. *arXiv*
310 *preprint arXiv:2503.10079*, 2025. URL <https://arxiv.org/abs/2503.10079>.

311 Yuan Liu, Haodong Duan, Yuanhan Zhang, Bo Li, Songyang Zhang, Wangbo Zhao, Yike Yuan, Jiaqi Wang,
312 Conghui He, Ziwei Liu, Kai Chen, and Dahua Lin. MMBench: Is your multi-modal model an all-around
313 player? In *European Conference on Computer Vision (ECCV)*, 2024. URL <https://arxiv.org/abs/2307.06281>.
314

315 Curtis G. Northcutt, Anish Athalye, and Jonas Mueller. Pervasive label errors in test sets destabilize machine
316 learning benchmarks. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and*
317 *Benchmarks Track*, 2021. URL <https://arxiv.org/abs/2103.14749>.

318 Fabio Rosenthal, Sebastian Schmidt, Thorsten Graf, Thorsten Bagdonat, Stephan Günnemann, and Leo
319 Schwinn. Unexplored flaws in multiple-choice VQA evaluations. *arXiv preprint arXiv:2511.22341*, 2025.
320 URL <https://arxiv.org/abs/2511.22341>.

321 Pratham Singla, Shivank Garg, Vihan Singh, and Paras Chopra. Do vision-language models see or guess?
322 measuring and reducing textual-prior reliance with a phrasing-controlled benchmark. *arXiv preprint*
323 *arXiv:2606.10400*, 2026. URL <https://arxiv.org/abs/2606.10400>.

324 Dingjie Song, Sicheng Lai, Mingxuan Wang, Shunian Chen, Lichao Sun, and Benyou Wang. Both text and images
325 leaked! a systematic analysis of data contamination in multimodal LLM. *arXiv preprint arXiv:2411.03823*,
326 2024. URL <https://arxiv.org/abs/2411.03823>.

327 Sang Truong, Yuheng Tu, Michael Hardy, Anka Reuel, Zeyu Tang, Jirayu Burapachee, Jonathan Perera,
328 Chibuike Uwakwe, Ben Domingue, Nick Haber, and Sanmi Koyejo. Fantastic bugs and where to find them in
329 AI benchmarks. *arXiv preprint arXiv:2511.16842*, 2025. URL <https://arxiv.org/abs/2511.16842>.

330 Joshua Vendrow, Edward Vendrow, Sara Beery, and Aleksander Madry. Do large language model benchmarks
331 test reliability? *arXiv preprint arXiv:2502.03461*, 2025. URL <https://arxiv.org/abs/2502.03461>.

332 Kaichen Zhang, Bo Li, Peiyuan Zhang, Fanyi Pu, Joshua Adrian Cahyono, Kairui Hu, Shuai Liu, Yuanhan
333 Zhang, Jingkan Yang, Chunyuan Li, and Ziwei Liu. LMMs-Eval: Reality check on the evaluation of large
334 multimodal models. *arXiv preprint arXiv:2407.12772*, 2024. URL <https://arxiv.org/abs/2407.12772>.

335 Zicheng Zhang, Xiangyu Zhao, Xinyu Fang, Chunyi Li, Xiaohong Liu, Xionghuo Min, Haodong Duan, Kai
336 Chen, and Guangtao Zhai. Redundancy principles for MLLMs benchmarks. In *Proceedings of the 63rd Annual*
337 *Meeting of the Association for Computational Linguistics (ACL)*, 2025. URL <https://aclanthology.org/2025.acl-long.612/>.
338

339 Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. Large language models are not robust
340 multiple choice selectors. In *International Conference on Learning Representations (ICLR)*, 2024. URL
341 <https://arxiv.org/abs/2309.03882>.

342 Runyu Zhou, Qi Zhang, Qixun Wang, and Yisen Wang. Diagnosing visual ignorance in vision-language models.
343 *arXiv preprint arXiv:2606.06890*, 2026. URL <https://arxiv.org/abs/2606.06890>.

A Reproducing every number in this paper

The audit consumes only the per-model result files that VLMEvalKit writes for a normal evaluation, plus the matching blind-run files. For each benchmark, the following two commands regenerate its statistics and figures; the HallusionBench invocation additionally passes `-design-prefix VD` to both scripts.

```
python scripts/audit_significance.py \
  -full <model>_MMStar_gpt-4o-mini_result.xlsx (one per model) \
  -blind <model>_MMStar_gpt-4o-mini_result_blind.xlsx \
  -models InternVL3-8B Ristretto-3B gpt-5-nano \
  -strata-column category -perms 999 -seed 7 -out <report dir>

python scripts/audit_paper_figures.py \
  -full ... -blind ... -models ... -stats-json ... -out
paper/figures
```

The blind artifacts themselves come from a normal evaluation invocation with the `-blind` flag, which is the only step that requires inference:

```
python run.py -data MMStar -model <model> -blind
```

The statistics script emits the canonical JSON report. The figure script reads all reported aggregate and stochastic values from that JSON and uses the evaluation files only for deterministic per-item panels; it does not rerun the permutation null. The scripts depend only on pandas/openpyxl (plus matplotlib for figures) and do not import the toolkit itself. Randomness enters only when the canonical report is generated, through the recorded permutation seed.

Answer extraction (exact arm). A prediction is mapped to an option letter by three ordered rules: a bare letter, possibly punctuated; a letter followed by a delimiter at the start of the string; or a letter following the words “answer” or “option”. Anything else becomes the abstention sentinel Z, which is excluded from consensus votes. This mirrors the toolkit’s own extractor, so the audit does not introduce a second, divergent notion of what a model answered.

Permutation null. For each of 999 permutations, one random permutation of question IDs is drawn and applied to the blind hit vectors of *all* models simultaneously. This preserves each model’s marginal blind accuracy and the correlation between models’ blind answers, and destroys only the per-question pairing between full and blind runs. The zero-gain share is recomputed on each permuted dataset with the identical category rule; the reported excess is observed minus null mean, and the one-sided p -value is $(g+1)/(N+1)$, where g counts permutations with a share at least as large as observed. Stratified nulls repeat the procedure within each category. For the split-half check, questions are randomly halved and the excess is computed in each half. Across 20 resplits, the median absolute difference between half-sample estimates is 1.5 pp on MMStar and 2.1 pp on HallusionBench; these values are stability diagnostics.

B Threshold sensitivity of the visual-dependent share

The visual-dependent share depends strongly on its definition, which is why the paper never headlines it. On MMStar (judged arm), the strict all-or-none definition flags 338 items (22.5%), while a majority definition (at least two of three models correct with the image, at most one without) flags 822 items (54.8%). On HallusionBench the same pair is 140 (14.7%) against 478 (50.3%). The text-only category has no threshold at $n = 3$, since the accuracy grid makes it exact equality, and the unanimous consensus count has none at any n ; those are the quantities the paper reports.

C Binary-format calibration check

Design labels are read from the question index prefix that HallusionBench itself assigns (VD for visual-dependent items, the remainder treated as visual-supplement), giving 591 positives and 360 negatives. The AUC is computed on the continuous per-question visual gain $a_q^F - a_q^B$, with ties counted as half

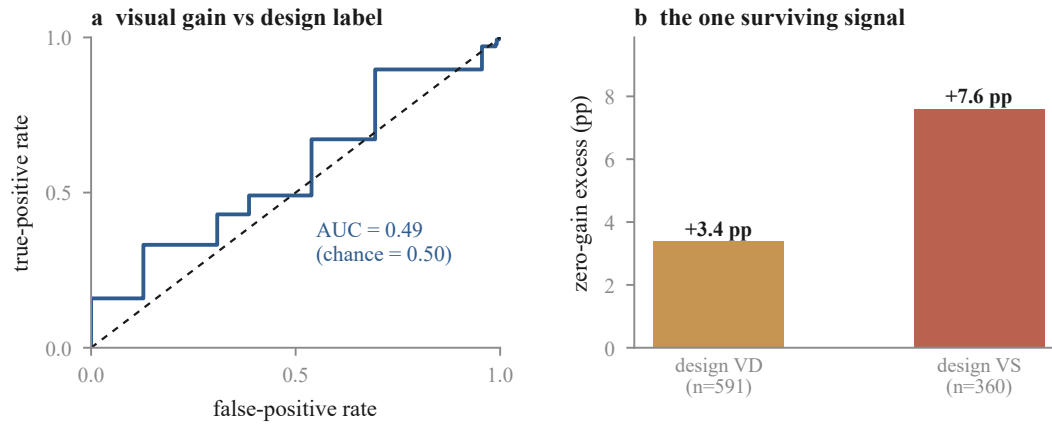


Figure 5: HallusionBench (951 yes/no items, same three models). **(a)** Ranking items by the continuous visual gain does not recover the benchmark’s own design labels (AUC 0.49 against 0.50 for chance). **(b)** The one directional signal that survives is consistent with the benchmark’s design: zero-gain excess is approximately 2.2 times larger on visual-supplement items than on visual-dependent items.

391 wins. Together with the degenerate letter-extraction behaviour described in Section 4.5, this is the
 392 evidence for excluding binary formats from the detectors’ current calibration.